THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 3: **The World of Metals and Non-metals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which statement is correct about carbon dioxide? 1

(a) it does not support burning **(b)** it supports burning
(c) it burns **(d)** it is a base

2. An object is dropped on the floor and gives a ringing sound. It is made of: 1

(a) sulphur **(b)** wood **(c)** metal **(d)** chalk

3. Which one produces a basic solution when dissolved in water? 1

(a) nitrogen dioxide **(b)** sulphur dioxide **(c)** calcium oxide **(d)** carbon dioxide

4. A basic oxide reacts with acid to form: 1

(a) gas and metal **(b)** salt and hydrogen
(c) salt and water **(d)** acid and water

5. A shiny black solid conducts electricity, is soft and is used as a pencil lead. It is: 1

(a) graphite **(b)** coal tar **(c)** iron **(d)** sulphur

6. Which oxide gives an acidic solution with water? 1

(a) sodium oxide **(b)** magnesium oxide **(c)** sulphur dioxide **(d)** calcium oxide

7. Which of the following is both a metal and a poor conductor of heat? 1

(a) lead **(b)** silver **(c)** copper **(d)** aluminium

8. Why does the green coating of copper form while aluminium forms a thin white layer? 1

(a) coating is paint **(b)** different reactions with air
(c) aluminium is liquid **(d)** copper is a gas

9. **Assertion (A):** The product formed on burning magnesium ribbon turns red litmus paper blue. 1

Reason (R): Oxides of metals are acidic in nature.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Metals react with dilute acids to release hydrogen gas. 1
Reason (R): Metals are good conductors of heat.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Explain why the copper vessel turns black when heated in air. 2

12. How can you show that the oxide of magnesium is basic using turmeric? 2

13. Why are iron nails coated with zinc or paint? 2

14. Why is mercury used in thermometers and barometers? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Why are cooking vessels made of steel and not of iron? 3

16. Why do overhead metal wires sag more in summer than in winter? 3

17. How is carbon dioxide used in fire extinguishers and why? 3

OR

Explain how you could separate a mixture of iron filings and sulphur powder using a magnet. 3

18. Why is a zinc coating given to iron buckets? 3

OR

What are alloys? Give two examples with components. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** Gurpreet places a clean iron nail in a blue copper sulphate solution. After about 30 minutes he sees a reddish-brown layer on the nail and finds that the blue colour of the solution has become pale green. 5
(i) What is the colour of the deposit on the nail?
(ii) What is the colour change in the solution?
(iii) Name this type of reaction.
(iv) Will copper displace iron if a copper strip is placed in iron sulphate solution? Give a reason.

20. Explain why metals conduct heat and electricity but non-metals generally do not, using only observations. 5

OR

Discuss the recycling of metals and its benefits for environment and economy. 5

END OF QUESTION PAPER — Total: 40 Marks